Forensics

Division C

Test Packet

UCSO

Science Olympiad

2025 University of Chicago Science Olympiad Invitational

Section 1: QUALITATIVE ANALYSIS

A. Powders [76]

Six different powder samples were found at the crime scene. Please identify each and fill out the table in the answer sheet. [42]

Supplemental Questions:

1. One of the reagents you have been provided with is hydrochloric acid. [8]
    a. What type of powder can HCl be used to detect? [1]

    b. Write out the full balanced equation for the reaction between powdered dolomite ($CaMg(CO_3)_2$) and hydrochloric acid. [4]

    c. Write out the full balanced equation for the reaction between aqueous sodium bicarbonate and hydrochloric acid. [3]

2. Which of the following is/are basic? [2]
    a. Sodium carbonate
    b. Calcium nitrate
    c. Sucrose
    d. Sodium acetate
    e. Calcium carbonate
    f. Cornstarch
    g. Ammonium chloride

3. Two powders are mixed together in a sample. Below are the sample's results after testing. [9]
    - HCl Added: Fizzes
    - NaOH Added: No reaction
    - pH: ~8
    - Benedict's Test: Blue
    - Iodine Added: Blue-black

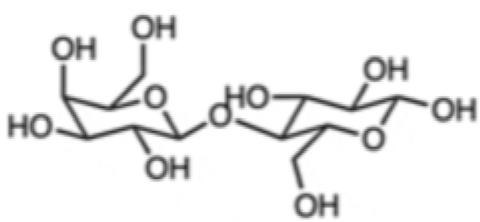
    a. Which two powder types are present (choose from list below)? [4]
        - Baking soda - Cornstarch - $CuSO_4$
        - Sugar Calcium carbonate - $CaCl_2$

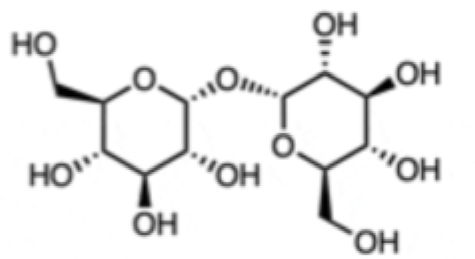
b. Would this sample be fully soluble in water? Why or why not? Explain what occurs at the molecular level when ionic compounds dissolve in water. [5]

4. The Benedicts test differentiates between reducing sugars and nonreducing sugars. [6]
   a. What is the difference between a reducing and nonreducing sugars? [2]

   b. Name another test that serves the same purpose. [2]

   c. Circle the reducing sugar below. [2]





5. Answer the following questions [9]
   a. What does it mean for a substance to be hygroscopic? Which of the powders found at the scene is hygroscopic? [2]

   b. Many hygroscopic solids contain ions with high charge density. Explain how charge density affects hygroscopic behavior at the molecular level. [3]

   c. A 5.00 g sample of anhydrous $MgSO_4$ is exposed to humid air and gains mass to become $MgSO_4{\cdot}7H_2O$. Calculate the theoretical final mass if the sample fully converts to the heptahydrate. [4]

B. Polymers [46]

Four different polymer samples were found at the crime scene. Due to an insufficient quantity of Polymers 4-6, you will not be able to physically test them. Investigators were, however, able to obtain descriptions of how they burn. Please identify each polymer and fill out the table in the answer sheet. [30]

Supplemental Questions:

1. What is a copolymer? [2]
2. What does the “6-6” designation mean in nylon 6-6? [2]
3. What is the scientific name of the polymer Teflon (abbreviation is fine). [2]
4. Describe two differences between condensation and addition polymerization. [4]
5. Which of the Forensics polymers is used to make shatterproof glass? [2]
6. Which of the Forensics polymers is used to make polyester? [2]
7. Draw the structure of the monomer that makes up polystyrene. [2]

C. Fibers [40]

Three different powder samples were found at the crime scene. Please identify each and fill out the following table [18]

Supplemental Questions:

1. Draw the cross sections of the following fibers. [3]
   a. Cotton
   b. Wool
   c. Silk

2. A fiber with a hollow core is observed. What advantage does this structure provide, and where is it commonly used? [3]

3. Answer the following questions regarding wool. [5]
   a. Is the reaction of wool absorbing water vapor exothermic or endothermic? [2]

   b. Why do acid dyes bind effectively to wool but poorly to cotton? [3]

4. A cotton fiber shows increased luster, enhanced dye uptake, and greater tensile strength. Explain how mercerization accounts for all three observations. [6]

D. Hair [46 points]

Five different hairs were found at the crime scene. You have been provided with images of each under a microscope. Please identify the species each hair came from and fill out the following table. [24]

Sample Number	Name (2)	Type of Cuticle (1)	Who it implicates (2)
1			
2			
3			
4			
5			

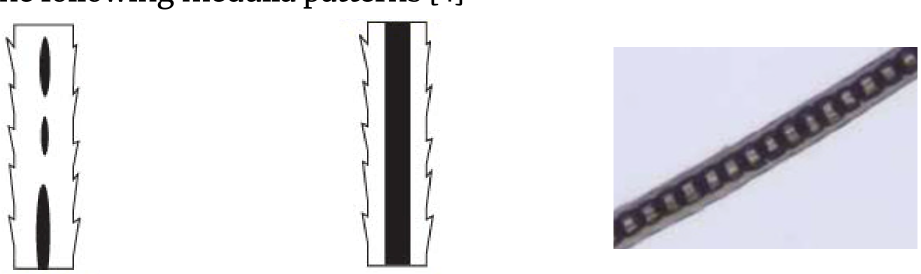
Supplemental Questions:

1. What are the three stages of hair growth, and roughly how long is a human hair in each? [3]

2. A questioned hair has a medullary index of 0.68, a continuous medulla, and thick pigmentation. Is this hair more likely human or animal? Give two reasons why. [3]

3. A hair found at a crime scene has a club-shaped root with no follicular tag. [4]
   a. What growth phase is this hair in? [2]
   b. What type(s) of DNA are likely present? [2]

4. Identify the following medulla patterns [4]

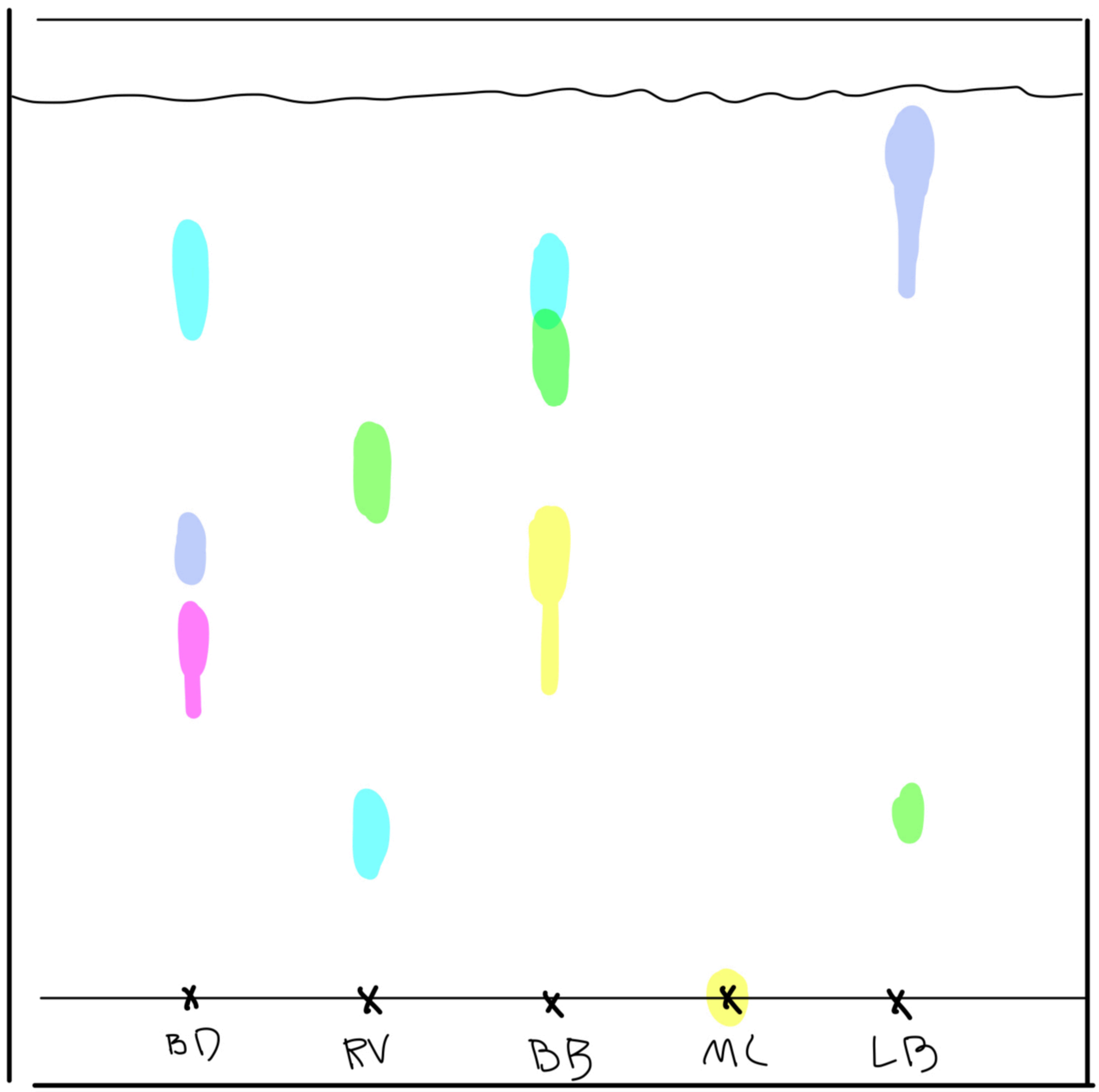


5. Melanin is the pigment responsible for hair color. [8]
   a. What are the two types of melanin found in human hair and which colors are they responsible for? [2]
   b. Which type provides UV protection? [1]

c. A red hair shows minimal pigment remaining after exposure to H_2O_2. What does this indicate about its melanin composition, and why? [2]
d. If a red hair remains red after exposure to H_2O_2, what might this indicate about the hair [3]

Section 2: CHROMATOGRAPHY AND MASS SPECTROMETRY [28]

Bobby takes a sample of ink from the crime scene note, along with ink from each suspect, and performs paper chromatography on them. He determines the Rf's of the crime scene ink to be 0.12 and 0.95.



1. What are the Rf's of each of the suspects' pens? [5, 1 pt each pen]

2. Which suspect is implicated by the chromatography? [2]

3. However, Bobby reminds his apprentices, Elise and Tom, that paper chromatography is not always the most accurate, and that they should use TLC when possible (they can't in this case because the lab is broke, rip). What is one reason that Bobby might have given Elise and Tom to support this preference for TLC over paper chromatography? [2]

4. Elise, inspired, attempts to perform TLC in her personal home lab (#rich) using the sample ink from the crime scene. However, when she checks on her TLC plate, she finds that her chromatogram has barely started developing. What mistake might she have made? (Select all that apply) [1]

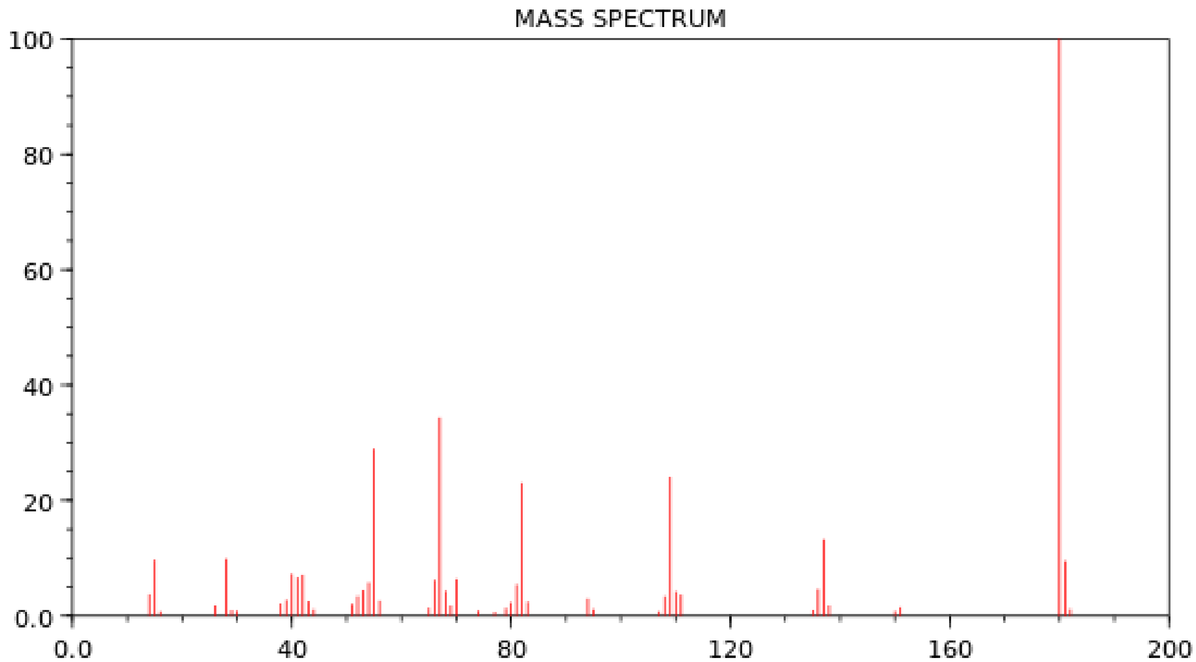
   a. Using hexane instead of ethyl acetate
   b. Leaving her TLC chamber uncovered
   c. Forgetting to purify her sample ink
   d. Using alumina instead of silica gel

5. Give one situation (be as specific as possible) when you might use normal-phase HPLC. Is reverse-phase or normal-phase HPLC more commonly used? [2]

6. Look again at MC's pen ink. Let's assume that dH_2O is used as the solvent.

   i. Propose a reason why the analyte has the Rf that it does. [1]
   ii. Predict which of these could be component in MC's ink could cause it to have this property. [2]
      a. Pencil graphite
      b. Sharpie
      c. Water-based dye
      d. Oil-based dye
      e. Starch-based dye
      f. Protein-based dye

The mysterious liquid was analyzed using mass spectrometry, and the results are shown below.



7. Tom forgot to label the axes of this mass spectrum. What are they? [2]

8. What is the molecular ion peak and base peak?[2]

9. What is the molar mass of this compound, in amu? [1]

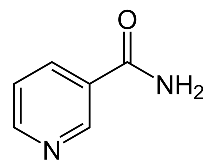
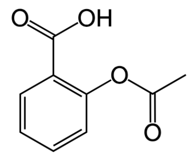
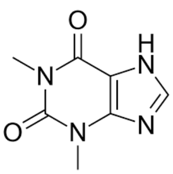
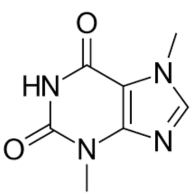
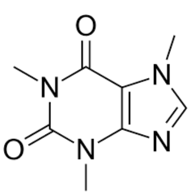
10. Explain the presence of the peak at 137 m/z, including the amu and identity of the fragment. [2]

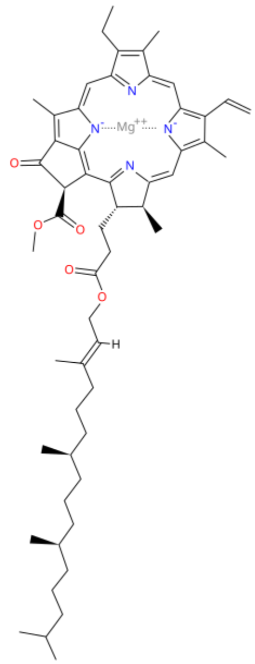
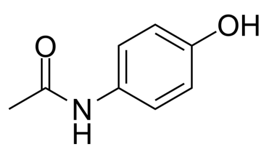
11. From this mass spectrum, how many nitrogen atoms are present in this compound? [1]

   a. 0
   b. 1
   c. 2
   d. 3
   e. 4
   f. 5
   g. 6

12. Identify the identity of the unknown compound, using the options below. [4]

Compound	Structure

Nicotinamide (Vitamin B3)	
Acetylsalicylic acid (Aspirin)	
Theophylline (used to treat COPD and asthma)	
Theobromine (found in chocolate)	
Caffeine	

Chlorophyll a	
Acetaminophen (Tylenol)	

13. Who does this incriminate, if anyone? [1]

Section 3: FINGERPRINTS [28]

The right index finger of the bloody handprint found on the broken window glass is shown below.

1. Define patent, latent, and plastic fingerprints, and determine which type was found here. [4]

2. Identify the fingerprint pattern, and who this incriminates. [2]

3. What are volar pads? What does the fingerprint pattern tell you about the nature of the volar pad for this finger? [2]

4. Approximately what percent of the population has this specific pattern of fingerprint? [1]
    a. 5%
    b. 15%
    c. 25%

d. 50%

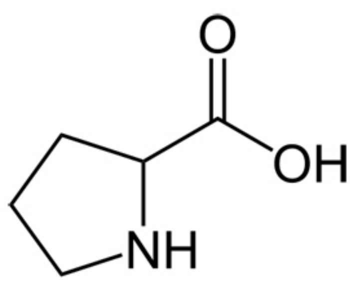
5. Which of these conditions can result in a lack of dermatoglyphics? (Select all that apply). [2]

a. Luminoderma fractalis
b. Naegeli-Franceschetti-Jadassohn syndrome
c. Larnell-Forsythe-Kadison syndrome
d. Dermatopathia Pigmentosa Reticularis
e. Keratochroma variegata

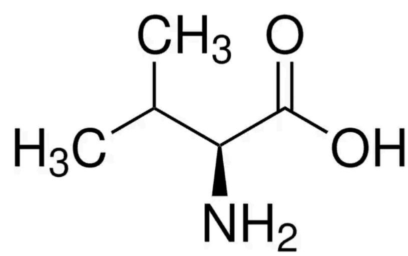
6. What color results from combining ninhydrin with the amino acid proline? [1]

7. What color results from combining ninhydrin with the amino acid valine? [1]

8. Explain the differences between your answers to 6 and 7 using the amino acid structures below. Make sure to include how the specific structural differences of proline and valine result in different ninhydrin products, and how that relates to color. [4]



Proline



Valine

9. What is the most common classification in the Henry System? [1]

10. Calculate approximately what percent of the population falls into the 1/1 primary classification category of the Henry Classification System. You must show your work to receive credit. [4]

11. However, what you calculated in (10) is not the true percentage of individuals with a 1/1 primary classification. Briefly explain why this is. Is the true percentage of individuals with a 1/1 primary classification higher or lower than wha you calculated in (10)? [2]

12. The IAFIS classification for a set of prints is, from 1–10:

LS	LS	WU	RS	WU	LS	AU	RS	LS	RS

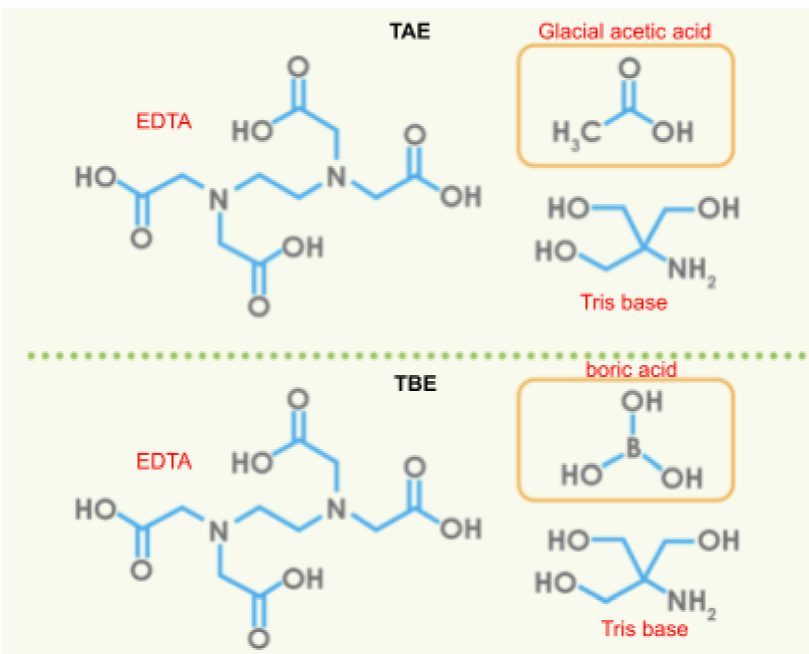
What would the Henry Classification number for this set of prints be? [4]

Section 4: DNA [27]

Bobby, the #professional forensics DNA analyst, decides to use this scandalous incident as a teaching experience for his two students, Elise and Tom. Each individual prepares to perform gel electrophoresis on agarose gel with DNA extracted from the root of **Hair 6**, along with DNA collected from each of the suspects.

1. Elise questions whether it is necessary to use TAE for the gel and buffer, and whether saving money can be achieved by using tap water instead. Name and explain two reasons why this is not a good idea. [4]

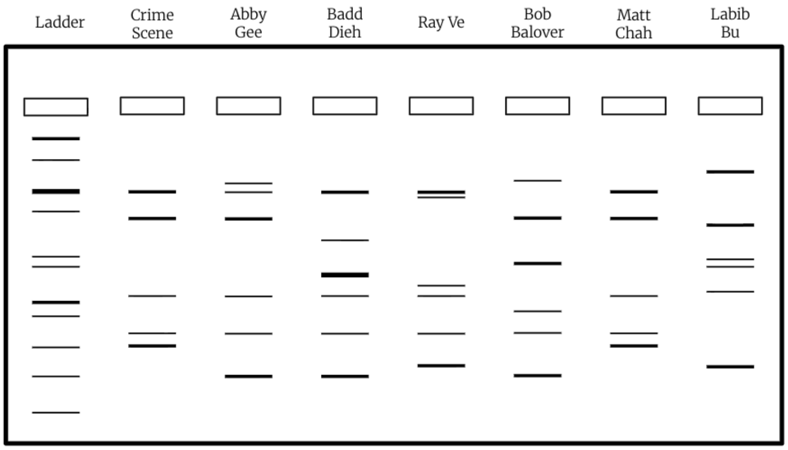
2. Tom, on the other hand, wants to be quirky and asks Bobby about using TBE instead of TAE. List one advantage of both TBE and TAE, and give an example of when you'd use each. You may use the diagram below as a reference. [4]



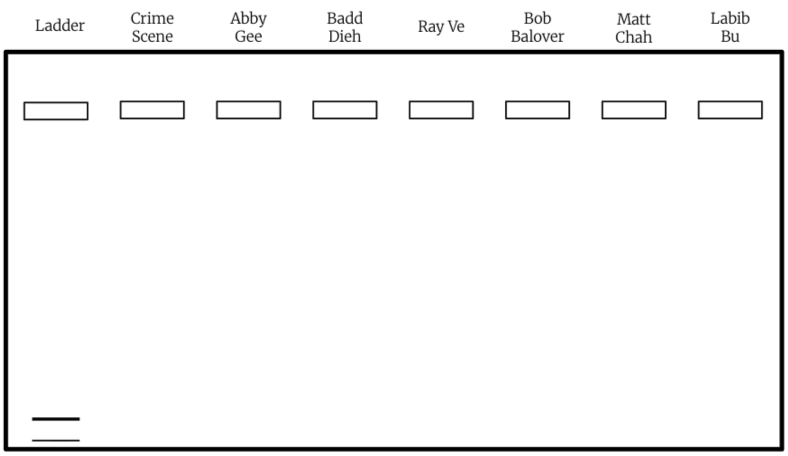
3. Given that the segments of DNA they're analyzing range from 500—2,000 bp, and their goal is DNA fingerprinting, identify and give an argument for which buffer (TAE or TBE) would be better to use. [2]

Bobby, Elise, and Tom, are finally able to prepare the correct buffers and gels, and each runs their gel electrophoresis at separate times.

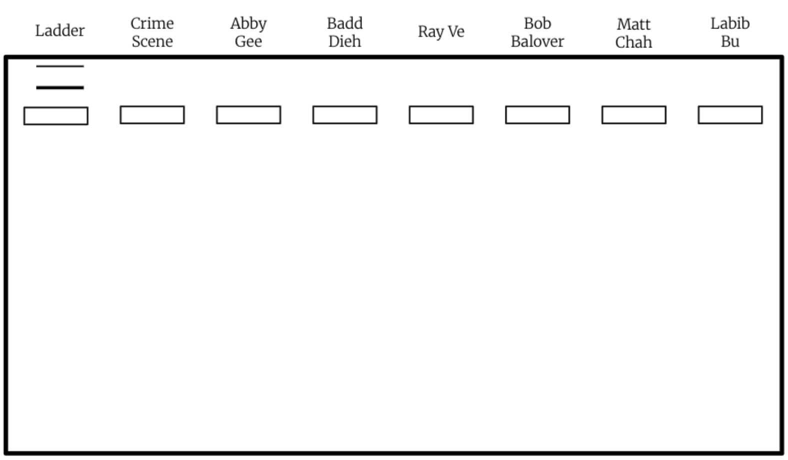
But, unsurprisingly, Elise's and Tom's gels don't look quite right at the end.



Bobby



Elise



Tom

4. What mistakes did Elise and Tom most likely make? [2]

5. From Bobby's gel, determine whose DNA was found at the crime scene. [1]

6. Fill in this table comparing PCR with *in vivo* DNA replication. Be as specific as you can when referring to the names of enzymes and temperature of reaction, when applicable. [6]

	In vivo DNA replication	PCR
What is the end result?	The entire genome is copied once per cell cycle.	
Step 1) Double-stranded DNA is separated		The PCR mix is heated to 96º C, which denatures the DNA strands by breaking hydrogen bonds.
Step 2) DNA replication is primed	RNA primers are synthesized against the single-stranded DNA by primase.	
3) Primers are extended, leading to double-stranded DNA	DNA polymerase adds dNTPs to the growing DNA strand.	Heat-resistant DNA polymerase adds dNTPs to the growing DNA strand at 72ºC.

7. Who won the Nobel Prize in Chemistry for inventing the PCR method, and in what year was this prize awarded? [1]

8. Fill in the blanks: The types are nucleotides used in Sanger Sequencing are________, and they differ from regular nucleotides because they lack ___________ on the __' carbon. [3]

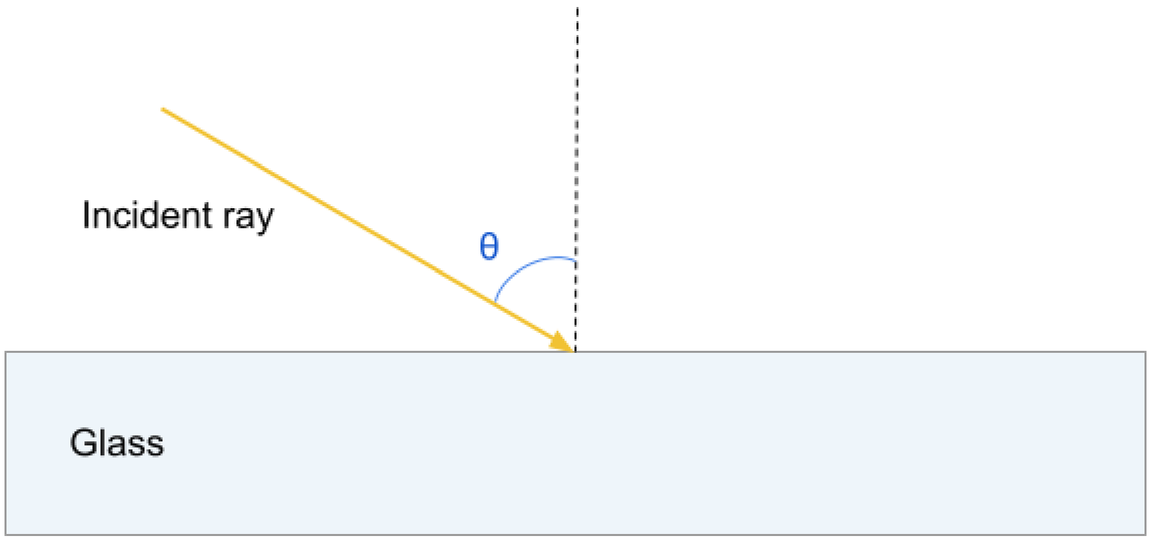
9. How many base pairs does the average human genome contain? [1]

10. Why do STR's have a relatively high mutation rate, and how does this affect DNA profiling? [3]

Section 5: GLASS AND BLOOD [32]

1. Pieces of glass were extracted from the suspects (they somehow all mysteriously had a broken glass incident within the last 24 hours) and analyzed. Finally, the type of glass found at the crime scene must be identified in order to use this evidence and pinpoint the criminal.

   a. Draw the path of the incident ray after entering the glass at an angle θ. [2]



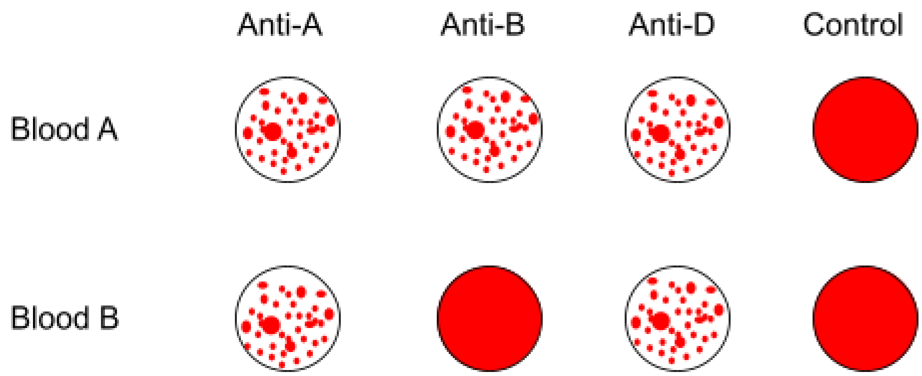
   b. Using $n_1 = refractive\ index\ of\ air\ (\sim 1)$, $x = the\ horizontal\ distance\ the\ light\ travels$, $y = the\ thickness\ of\ glass$, and $\theta = angle\ of\ incident\ ray\ to\ the\ normal$, determine the refractive index of the glass. [3]

2. Given the conditions θ = 51.4°, x = 3.00 cm, and y = 5.00 cm, determine the refractive index of the glass. Report your answer to three significant figures. [2]

3. What type of glass is this? Who does this incriminate? [2]

4. Bobby gives Elise and Tom the challenge of determining the refractive index of the window glass, using only an array of liquids with known refractive indexes and a microscope. Is this a possible feat? Explain your answer. [2]

Both **Blood A** and **Blood B** from the crime scene were analyzed with an antibody test. The results are shown in the image below.



5. Fill in the table below. [4]

	Blood type	Whose blood does this blood type match?
Blood 1		
Blood 2		

6. What blood type(s) can the people of the indicated blood type donate to? [2]

7. What might happen if someone is given a blood transfusion of the wrong blood type? [1]

8 If a spatter is 4.49 mm wide and 4.88 mm long, what is the angle of impact (to two significant figures)? At what speed was this blood spatter traveling (give a range)? [2]

9. What are the components of human blood? Write your answer in order of <u>least</u> to <u>most</u> prevalent. [4]

10. Blood type is dependent on genetic inheritance. A person with type B+ blood had children with a person with type AB- blood. What blood types are possible among their children? Account for every possibility of parental genotypes and assume no linkage. [2]

11. Who discovered the ABO blood group system, and in what year? [2]

12. The Kastle-Meyer test is a chemical test used in forensics to detect the possibly presence of blood.

a. What reagent is reduced in the Kastle-Meyer test to produce the color change in a positive test result, and what is this color? [2]

b. For which of these substances can the Kastle-Meyer test produce a positive test result? [2]
    i. Carbon monoxide
    ii. Cobalt
    iii. Hemogloblin
    iv. Horseradish
    v. Iron
    vi. Manganese
    vii. Plastic
    viii. Potato
    ix. Saliva
    x. Sugar
    xi. Tears
    xii. Turmeric
    xiii. Vegetable oil
    xiv. Water

c. Is the Kastle-Meyer test a presumptive or confirmatory test for the presence of blood? What characteristic of the test makes it presumptive or confirmatory? [2]

Section 6: ANALYSIS [106]

Please write your analysis where indicated on the answer sheet.